The Intrinsic Dimensionality of the Stock Market: A Geometric Approach to Factor-Based Market Analysis

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by

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CERTIFICATE

*This is to certify that the work contained in this thesis entitled "**The Intrinsic Dimensionality of the Stock Market: A Geometric Approach to Factor-Based Market Analysis**" is a bonafide work of **Rohan Kumar Mahato** (**Roll No. 220101088**), carried out in the Department of Computer Science and Engineering, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.*

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Chapter 1

Introduction

The Indian equity market with broad indices like the NIFTY 500 is kind of a high dimensional system with many securities. Actually for someone doing quantitative work this feels quite hard because keeping track of more than 500 companies is very heavy to compute and many times the stats become not so reliable and there is lot of noise. In this thesis i try to look into this problem by using a more geometric way for factor based study. Instead of only thinking of the usual statistical correlations we check out how the data looks in high dimensional geometry. This way we try to see what is the real dimension inside teh market and which few factors are actually driving the price movements in this big cloud of data.

1.1 The Challenge: The "Curse of Dimensionality"

The huge amount of data in the NIFTY 500 makes something people call the curse of dimensionality. In this type of high dimension it becomes tough to separate proper market signals from random noise when using normal statistical methods. When the number of dimensions or stocks keeps increasing the data gets more sparse and the distance ideas that we use for clustering or correlation stop giving that clear meaning.

For an analyst checking the covariance of 500 assets means calculating $\frac{500\times499}{2}$ different pairwise correlations. This big increase in computation makes the system sensitive. Even a small noise in the data can change the risk models a lot. also the geometric understanding we have in two or three dimensions does not really work here. As we see in later chapters high dimensional vectors act in strange ways like distance concentration and a kind of built in orthogonality which we need to think about to make strong market models.

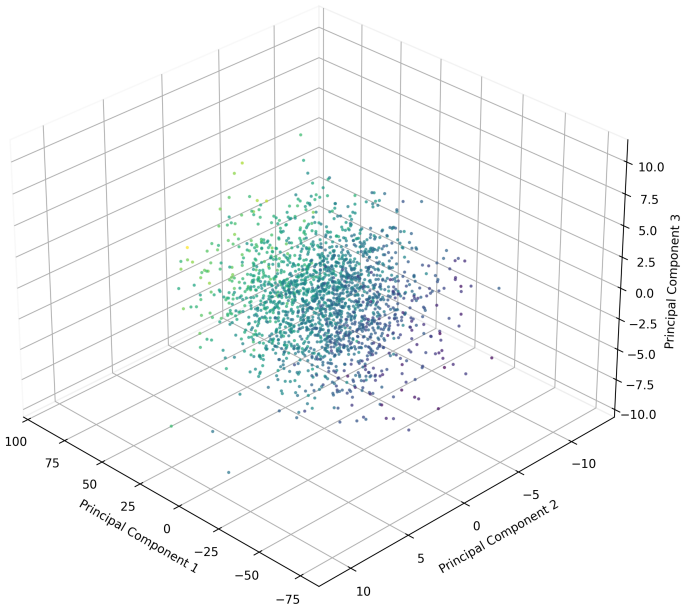


Fig. 1.1 Visualization of a simulated high-dimensional market environment projected into 3D space. Each point represents the collective returns of hundreds of securities on a single day, showing how hard it is to track co-movements in the raw data cloud.

1.2 Research Objectives and Methodology

The main problem we look at in this thesis is the analytical complexity of this high dimensional market space. The main objective is to make and test a method that can reduce the dimension of the market study. So we want to go from a space with hundreds of companies to a much smaller space of important hidden factors.

By using dimensionality reduction and also using the geometric properties of high dimensional spaces we try to:

- find the intrinsic dimension k of the NIFTY 500.
- make a reduced dimension model that can explain the primary drivers of the Indian stock market.
- tell apart the systematic market modes and random noise.

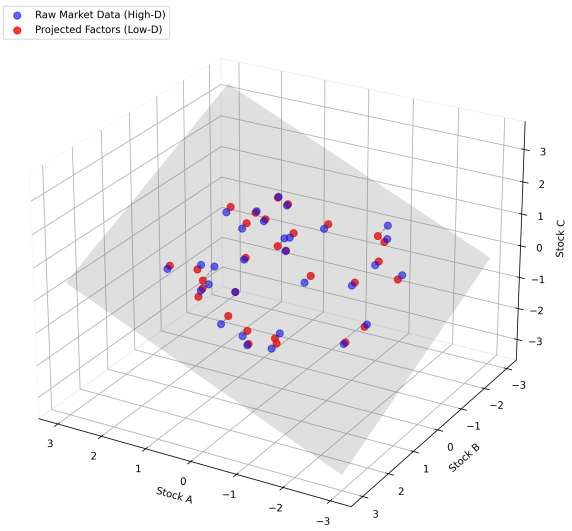


Fig. 1.2 Conceptual illustration of projecting high-dimensional market data onto a lower-dimensional subspace to find hidden factors.

1.3 Organization of The Report

The rest of this thesis is organized to slowly connect the theory of high-dimensional geometry with how it can be actually used for analyzing financial markets.

Chapter 2 (Motivation) looks at the limits of normal statistical methods when we have very high-dimensional data. It explains the main geometric idea and why we need to move from a high-dimensional "Stock Space" to a smaller "Factor Space."

Chapter 3 (The Geometry of High-Dimensional Spaces) gives the theory behind the research. It shows the weird properties of high-dimensional vectors, like the Law of Large Numbers, the Concentration of Measure, and the Gaussian Annulus Theorem.

Chapter 4 (Best-Fit Subspaces and SVD) gives the math for reducing dimensions. It explains how Singular Value Decomposition (SVD) works and talks about the theoretical guarantees from the Johnson-Lindenstrauss Lemma.

Chapter 5 (Financial Data as Geometric Vectors) connects theory with actual finance. It explains the Vector Space Model for stock returns and compares real market data with a random Gaussian distribution.

Chapter 6 (Methodology) shows how we actually do the analysis. It explains the data pipeline for NIFTY 500, including log-return calculation, standardization, and steps for subspace projection.

Chapter 7 (Expected Outcomes) explains what we expect from the factor analysis. It talks about Scree plots to find the intrinsic dimension and how to separate real market signals from random noise.

Finally, **Chapter 8 (Conclusion and Future Work)** sums up the results of the geometric approach and shows the plan for Phase 2 of the project, including testing out-of-sample and doing more detailed sector analysis.

Chapter 2

Motivation

2.1 The Analysis Paradox

Modern financial markets have a lot of data. The NIFTY 500 with the top 500 companies in India keeps giving a huge flow of price updates which makes a high dimensional dataset that looks like a goldmine for any analyst. But actually this big amount creates a kind of paradox. When the dimension of the data becomes too high the normal statistical methods stop giving clear insights. People call this the Analysis Paradox or the Curse of Dimensionality.

In a 500 dimensional space the data turns quite sparse. The usual distance metrics that we use for clustering similar stocks or finding correlations stop being very meaningful. If someone treats daily market returns as only random variables then the geometric distance between any two days becomes almost same when the dimensions increase. But financial sense tells us it is not like that. Markets have rallies and crashes and sector based moves. This mismatch between the high dimensional statistical theory and the real market behaviour is the main motivation for this thesis.

2.2 From Stock Space to Factor Space

The limits of studying 500 individual covariance pairs make us shift our view. So we move from Stock Space where every company is a dimension to Factor Space.

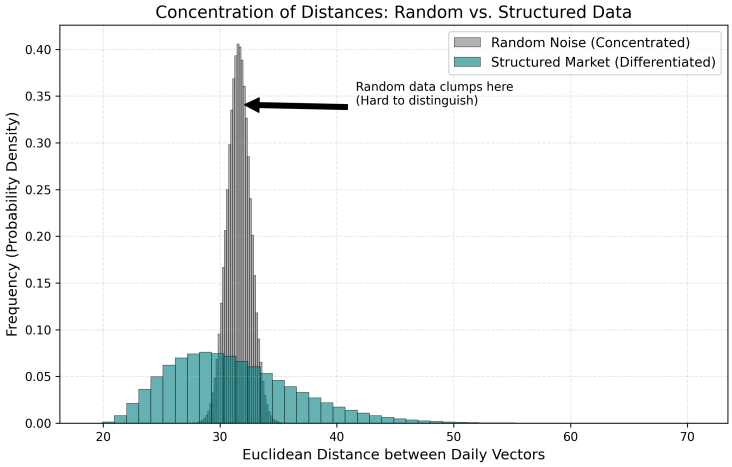


Fig. 2.1 Histogram showing the concentration of distances in high dimensional space. In a purely random market the distances would get concentrated in a narrow range which makes differentiation very tough. Real markets show some structure.

The motivation is that the Indian market is not moved by 500 separate forces but by a much smaller group of hidden factors like global market feeling, inflation or sector performance. By using high dimensional geometry we can project the market into this lower dimensional subspace without losing the important signal. This idea which is based on the theory by Blum, Hopcroft and Kannan [BHK20] helps us separate the systematic risk or signal from the idiosyncratic volatility or noise.

Chapter 3

The Geometry of High-Dimensional Spaces

3.1 Introduction

High dimensional data study is quite important for understanding complex systems like financial markets. But the geometry of high dimensional space R^d is very different from the two or three dimensional spaces that match our normal thinking. As mentioned in some basic literature [BHK20] if we make random points in d dimensions where each coordinate is a zero mean and unit variance Gaussian then some strange properties start showing up.

Two properties are very important for this thesis:

1. **The Concentration of Measure:** When d is large the Euclidean distances between almost all pairs of random points become nearly same with high probability.

2. **Volume Concentration:** The volume of a high dimensional unit ball is not spread everywhere. It mostly sits near the surface or the thin shell and also near its equator.

These things show that if the market was fully random then stock return vectors would all look almost same. If we see some structure then it means the data is in some lower dimensional manifold.

3.2 The Law of Large Numbers and Distance Concentration

The math reason for why distances concentrate comes from the Law of Large Numbers or LLN. Take two points $\mathbf{y}$ and $\mathbf{z}$ in a high dimensional space. The square distance between them is:

$$|\mathbf{y} - \mathbf{z}|^2 = \sum_{i=1}^{d} (y_i - z_i)^2 \tag{3.1}$$

We can think of this as the sum of d independent samples of a random variable $x_i = (y_i - z_i)^2$. Since the coordinates have bounded variance the LLN works here.

The LLN says that with high probability the average of these samples will move close to the expected value of the random variable. The chance that the average changes from the expectation $E(x)$ by more than ϵ is given by Chebyshev's inequality:

$$\text{Prob}\left(\left|\frac{x_1 + \cdots + x_n}{n} - E(x)\right| \geq \epsilon\right) \leq \frac{Var(x)}{n\epsilon^2} \tag{3.2}$$

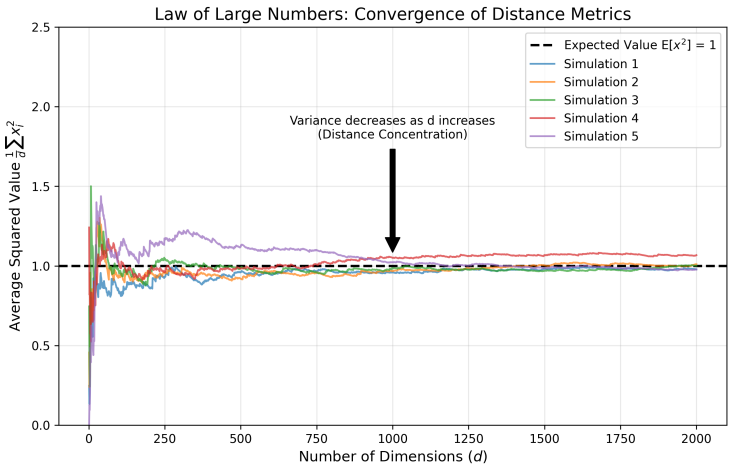


Fig. 3.1 Convergence graph showing the Law of Large Numbers. When the number of dimensions increases the observed average gets tighter around the expected value which causes distance concentration.

When the dimension d or n grows large the n in the denominator becomes big and pushes the probability close to zero. This forces the distance to concentrate around one

value. In financial terms this means that in a high dimensional isotropic market purely random return vectors would all look equally far from each other which makes clustering or separation not possible.

3.3 Volume Concentration and The Thin Shell

A key property of high dimensional objects is that most of their volume stays near the surface. Take any object A in R^d. If we shrink A by a small factor ϵ to make a new object $(1-\epsilon)A$ then the volume changes with the dimension as:

$$\text{volume}((1-\epsilon)A) = (1-\epsilon)^d \text{volume}(A) \tag{3.3}$$

Using the inequality $1 - x \leq e^{-x}$ we get:

$$(1-\epsilon)^d \leq e^{-\epsilon d} \tag{3.4}$$

When we fix ϵ and let $d \to \infty$ this number goes to zero very fast. This shows that almost all the volume of A must be sitting outside the inner core $(1-\epsilon)A$.

For a unit ball this means that at least $1 - e^{-\epsilon d}$ fraction of the volume stays in a small annulus of width ϵ near the boundary. In simple words in high dimensions the inside of the ball is practically empty and all points sit on a thin shell near the boundary.

3.4 The Gaussian Annulus Theorem

For modeling stock returns we usually assume a Gaussian distribution. The ideas of volume concentration also work for Gaussian distributions through the Gaussian Annulus theorem.

For a d dimensional spherical Gaussian with unit variance in each direction the expected squared distance from the center is d. So the radius of the Gaussian is $\sqrt{d}$. The theorem says that the points are tightly packed around this radius.

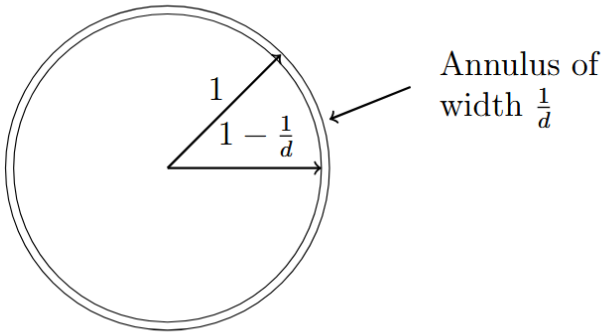


Fig. 3.2 Illustration of the Thin Shell idea. Most of the volume of the d dimensional ball of radius r stays in an annulus of width $O(r/d)$ near the boundary.

Theorem: For a d dimensional spherical Gaussian with unit variance and for any $\beta \leq \sqrt{d}$ all but at most $3e^{-c\beta^2}$ of the probability mass stays in the annulus:

$$\sqrt{d} - \beta \leq |\mathbf{x}| \leq \sqrt{d} + \beta \tag{3.5}$$

where c is some positive constant.

This theorem is very important for our study. It gives a strict baseline for randomness. If the NIFTY 500 data was only random noise then the daily return vectors would stay exactly in this thin annulus at radius $\sqrt{d}$. If we see any deviation like points going far outside this annulus during a crash or points clustering in one zone during sector rallies then it is a math proof that the market has real structure and lower dimensional behaviour.

Chapter 4

Best-Fit Subspaces and Singular Value Decomposition (SVD)

4.1 Introduction: The Geometry of Best Fit

In this chapter we look into the Singular Value Decomposition or SVD of a matrix which acts like the main tool for our market factor study. Think of the market data matrix A of size $n \times d$ where each row shows one day's market state as a point in d dimensional Stock Space.

The aim of SVD is to find the best fitting k dimensional subspace for this cloud of n points. In the basic text best has a double meaning because of the Pythagorean Theorem:

1. **Minimizing Geometric Error:** reducing the sum of squared perpendicular distances of the points to the subspace.

2. **Maximizing Information Retained:** increasing the sum of squared lengths of the projections of the points on the subspace.

This double idea is very important. By finding the subspace that gives maximum projection lengths we find the directions or market factors that capture the maximum variance or information of the NIFTY 500.

4.2 Singular Value Decomposition (SVD)

In matrix form the Singular Value Decomposition of a real matrix A is written as:

$$A = UDV^T \tag{4.1}$$

Where:

- The columns of U (size $n \times r$) are orthogonal unit vectors called the **left singular vectors**. In our case these show how factors change over time.
- The columns of V (size $d \times r$) are orthogonal unit vectors called the **right singular vectors**. These show the portfolio weights of the stocks for each factor.
- D or Σ is a diagonal matrix with positive entries $\sigma_1, \sigma_2, \ldots, \sigma_r$ known as the **singular values**.

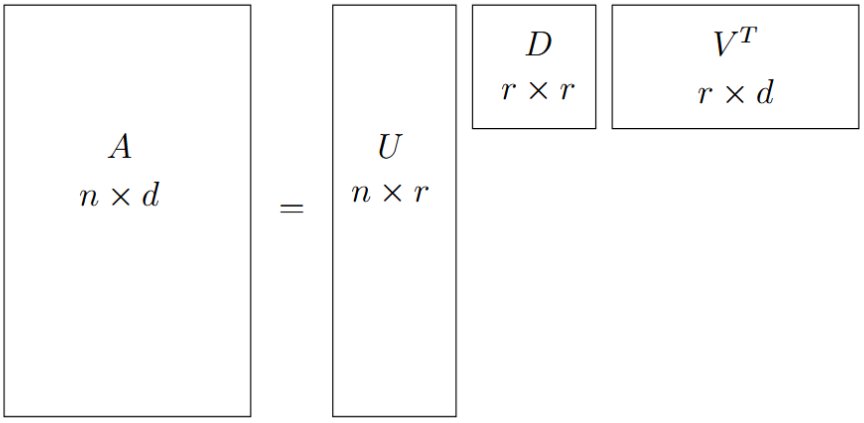


Fig. 4.1 The SVD decomposition of an $n \times d$ matrix into orthogonal pieces U and V scaled by diagonal matrix D.

4.2.1 Rank-One Decomposition

The matrix equation can also be shown as a sum of rank one matrices:

$$A = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^T \tag{4.2}$$

Here the ith term $\sigma_i \mathbf{u}_i \mathbf{v}_i^T$ is the ith principal component of the market. The singular value σ_i tells the strength of this component. Since we keep them in decreasing order so $\sigma_1 \geq \sigma_2 \geq \ldots$ the first component shows the main market trend or Market Mode and later ones show other independent systematic changes.

4.3 The Greedy Algorithm for Singular Vectors

The SVD can be made using a greedy method. First we find the single line through the origin that fits the data best. Let $\mathbf{v}_1$ be the unit vector on this line. The first singular value is $\sigma_1(A) = |A\mathbf{v}_1|$.

For the next components we find the best fit line that is perpendicular to all earlier lines. The kth singular vector is:

$$\mathbf{v}_k = \arg \max_{\substack{\mathbf{v} \perp \mathbf{v}_1, \ldots, \mathbf{v}_{k-1} \\ |\mathbf{v}|=1}} |A\mathbf{v}| \tag{4.3}$$

This makes sure each found market factor is not correlated with the others so we can separate different market drivers like seeing a Tech Sector rally apart from a general Market rally.

4.4 Random Projections and the Johnson-Lindenstrauss Lemma

One main question in this thesis is: *Can we reduce 500 dimensions to a small k without messing up the market structure?* The theory that answers this is the Johnson Linden-

strauss or JL Lemma.

Theorem (Johnson-Lindenstrauss Lemma): For any $0 < \epsilon < 1$ and any set of n points in R^d a random projection $f : R^d \to R^k$ keeps the pairwise distances safe if $k \geq \frac{3}{c\epsilon^2} \ln n$. specifically:

$$(1-\epsilon)\sqrt{k}|\mathbf{v}_i - \mathbf{v}_j| \leq |f(\mathbf{v}_i) - f(\mathbf{v}_j)| \leq (1+\epsilon)\sqrt{k}|\mathbf{v}_i - \mathbf{v}_j| \tag{4.4}$$

This holds with probability at least $1 - \frac{3}{2n}$.

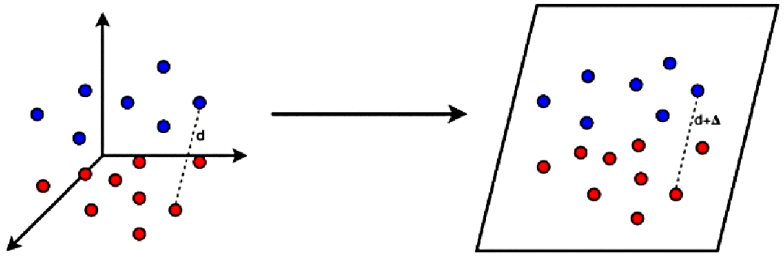


Fig. 4.2 Visualizing how relative distances stay safe under projection. The JL Lemma says the market shape stays almost same even after going to a much lower dimension.

The Remark in Theorem points out that the target dimension k depends on the log of the number of points n and not on the original dimension d. This gives the theory reason why a market with thousands of stocks can still be understood using only a few factors.

Chapter 5

Financial Data as Geometric Vectors

5.1 The Vector Space Model: From Text to Tickers

The main idea used in this thesis actually comes from the old work of Salton, Wong, and Yang (1975) in Information Retrieval [SWY75]. They showed how to represent text documents as vectors in a big space where each word acts like one dimension. This moved things from strict Boolean rules to simple linear algebra. It also made it easy to check similarity by looking at angles between vectors.

Here we try to use the same idea for finance. We think of the NIFTY 500 not like many separate time-series. Instead we think of it like one geometric object.

- **Dimensions (N):** Each stock becomes one axis in the space.
- **Points (T):** The full market on a day t becomes one point (vector) in this N-dimensional place.

Mathematically, the market state vector $\mathbf{r}_t$ for day t is defined as:

$$\mathbf{r}_t = [r_{1,t}, r_{2,t}, \ldots, r_{N,t}]^T \tag{5.1}$$

where $r_{i,t}$ is just the standardized log-return of stock i. When we look at the market like

this, the correlation between assets basically becomes the geometric alignment between two vectors (cosine similarity).

5.2 The Null Hypothesis: Random Matrix Theory

To show that the market actually has low-dimensional structure, we first need to say what a market without any structure looks like. This idea comes from the Econophysics literature, especially the work of Laloux et al. (1999) on Noise Dressing [LCBP99].

Laloux used Random Matrix Theory (RMT) on financial correlation matrices. The main point is that when the number of assets N is big compared to the time length T, the correlation matrix gets filled with noise. If the market was fully random (the Null Hypothesis), then the eigenvalues would follow the Marchenko-Pastur law. Almost all eigenvalues would sit in one bulk that shows noise. It would make a type of geometric hypersphere where no direction stands out too much.

5.3 Universal vs. Non-Universal Properties

Adding to Laloux's findings, Plerou, Gopikrishnan, et al. (1999) explained the difference between universal properties of complex systems (noise) and non-universal ones that belong only to the market (signals) [PGR+99].

In their study of financial cross-correlations, they found that around 94% of the eigenvalue spectrum follows random noise. But some parts do not fit into that bulk.

1. **The Market Mode:** The biggest eigenvalue (λ_{max}) is much larger than what RMT allows. Plerou et al. used the Inverse Participation Ratio (IPR) to show that this eigenvector is delocalized (it affects almost all stocks together). Geometrically it is the direction where the data cloud stretches the most.

2. **Sectoral Clusters:** Smaller deviations from the RMT bulk match with industry sectors like IT or Banking. These give secondary geometric axes.

5.4 Empirical Evidence: Fat Tails and Structure

The geometric idea depends on the fact that real market data is not a perfect sphere. It is more like an ellipsoid. Basic finance theory sometimes assumes Gaussian returns. But (as shown in Figure 5.1) real data has Fat Tails and a higher center.

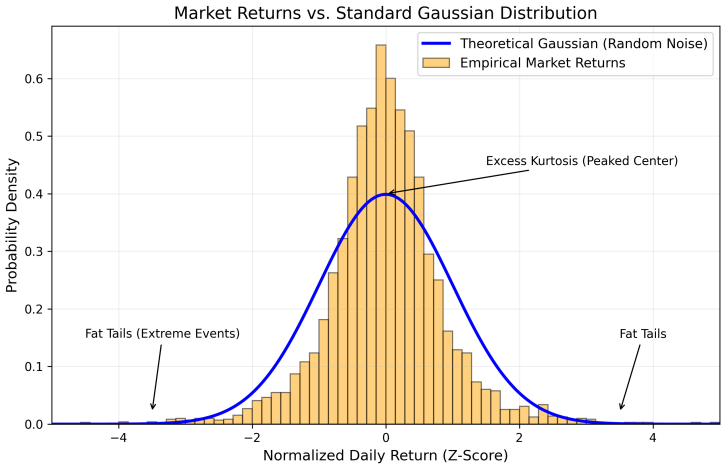


Fig. 5.1 Comparison of Empirical Market Returns vs. Theoretical Gaussian Distribution. The empirical data (orange) shows a higher peak and heavier tails than the Gaussian baseline (blue). This statistical deviation is the physical evidence that the market is not a random hypersphere, but a structured geometric object susceptible to dimensionality reduction.

The fact that the real distribution spills out beyond the Gaussian curve (the tails) and also peaks higher in the center shows what Plerou and Laloux said. The market does have non-random structure. This makes it possible to ignore the spherical noise (the bulk) and then project the data onto the few stretched axes (the factors) using SVD. It basically helps us separate the useful signal from the noise.

Chapter 6

Methodology: Subspace Projection of the NIFTY 500

6.1 Data Collection and Preprocessing

The base of this geometric analysis is a solid dataset for the Indian equity market. The data pipeline is made to turn raw price signals into a cleaner and more stable form (so it fits nicely in the geometric setup).

1. **Data Source:** We use historical daily adjusted closing prices for all stocks in the NIFTY 500 index for around 10 years. I picked this bigger index instead of the NIFTY 50 because it actually shows more high-dimensional effects between large, mid and small-cap stocks.

2. **Logarithmic Returns:** Raw prices (P_t) keep drifting and are not good for geometric work. So we change them into logarithmic returns:

$$R_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{6.1}$$

   Log-returns are better since they add over time and look more symmetric. It makes

them good for Euclidean space vectors.

3. **Z-Score Standardization:** To stop very high-priced stocks (like MRF) from taking over the variance, we standardize each stock's return series to have mean zero and variance one:

$$z_{i,t} = \frac{R_{i,t} - \mu_i}{\sigma_i} \tag{6.2}$$

This step puts all 500 stocks on the same level. It helps SVD focus on correlation structure instead of price size.

6.2 Matrix Construction and Factorization

We make a data matrix X of size $T \times N$, where T is the trading days and N is around 500 securities.

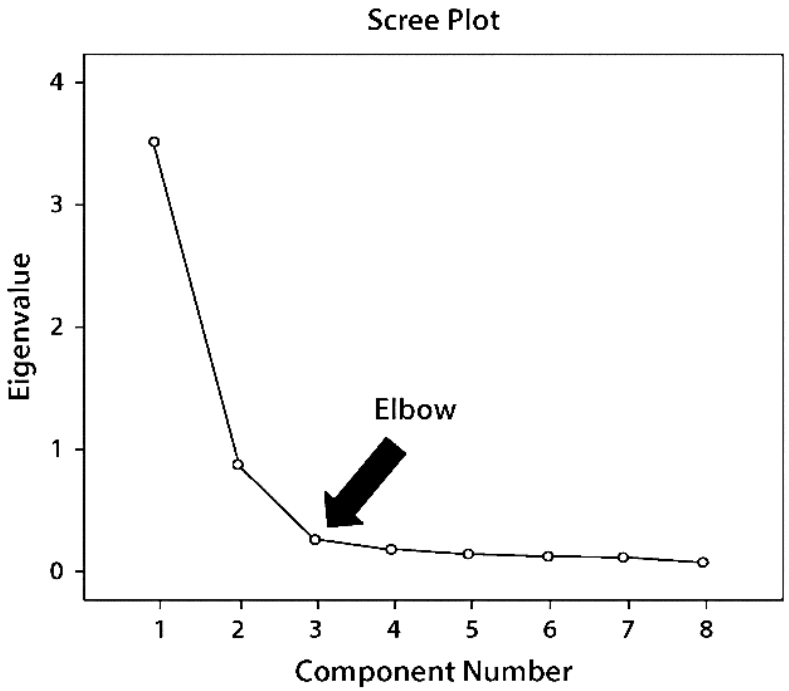


Fig. 6.1 Scree Plot of the Eigenvalues. The sharp drop in the first few components represents the 'Signal' (Market and Sector modes), while the flattened tail represents the 'Noise Floor' predicted by Random Matrix Theory.

To pull out the hidden factors, we use Singular Value Decomposition (SVD) straight on the correlation matrix $C = \frac{1}{T-1} X^T X$. Based on the method from Kumar et al. (2020) [KKK20], the eigenvalues help us split signal and noise.

6.2.1 Intrinsic Dimension Determination

The main step is to find the intrinsic dimension k (the number of factors we keep). We use the **Scree Plot** approach where we plot eigenvalues λ_i against the component index i.

We choose the intrinsic dimension k at the elbow of the plot where eigenvalues clearly move away from the random part. This makes it easy to drop the noise bulk and rebuild the market using only the k components that have real economic info.

Chapter 7

Expected Outcomes and Market Factor Analysis

7.1 Dimensionality Reduction Results

The main expected outcome of this research is to actually show in numbers that the NIFTY 500, even though it has 500 return series, really has a much smaller intrinsic dimension. Based on the work by Plerou et al. [PGR+99] and Sood et al. [SB20], we think the intrinsic dimension k will be somewhere between $10 \leq k \leq 20$.

This shows a big compression of information:

$$\frac{k}{N} \approx \frac{15}{500} = 3\% \tag{7.1}$$

So basically around 97% of the geometric degrees of freedom in the market come from noise or idiosyncratic risk. Only around 3% show the main systematic forces that push the Indian economy.

7.2 Interpreting the Geometric Factors

The subspace projection does not just shrink the data. It also separates different economic forces. The Singular Value Decomposition ($A = U\Sigma V^T$) gives a direct meaning for each geometric component.

- **The Market Mode (Component 1):** The first eigenvector ($\mathbf{v}_1$) matches with the biggest singular value. We expect this vector to be delocalized which means it has positive values for almost all stocks ($v_{1,i} > 0$ for all i). In geometric terms, this is like the diagonal of the hypercube where the whole market moves together. In finance it becomes the Market Beta and it shows broad macro feeling (like reaction to GDP numbers or RBI rate changes).

- **Sectoral Modes (Components 2 to k):** Since the singular vectors are orthogonal ($\mathbf{v}_2 \perp \mathbf{v}_1$), the next components catch systematic moves that go against the main market trend. The second component usually separates big sectors such as Defensive vs Cyclical stocks or sometimes IT vs Banking. These factors help us see how capital moves inside the market.

7.3 Visualization and Clustering

By projecting the 500 dimensional data on the plane made by the first two principal components (PC_1 and PC_2), we can see the market structure in a simple plot.

In Figure 7.1, stocks from the same industry (like TCS, Infosys, Wipro) usually gather close together in the reduced space even though the algorithm does not know their labels. This shows that the geometric method actually caught the basic economic similarities between companies.

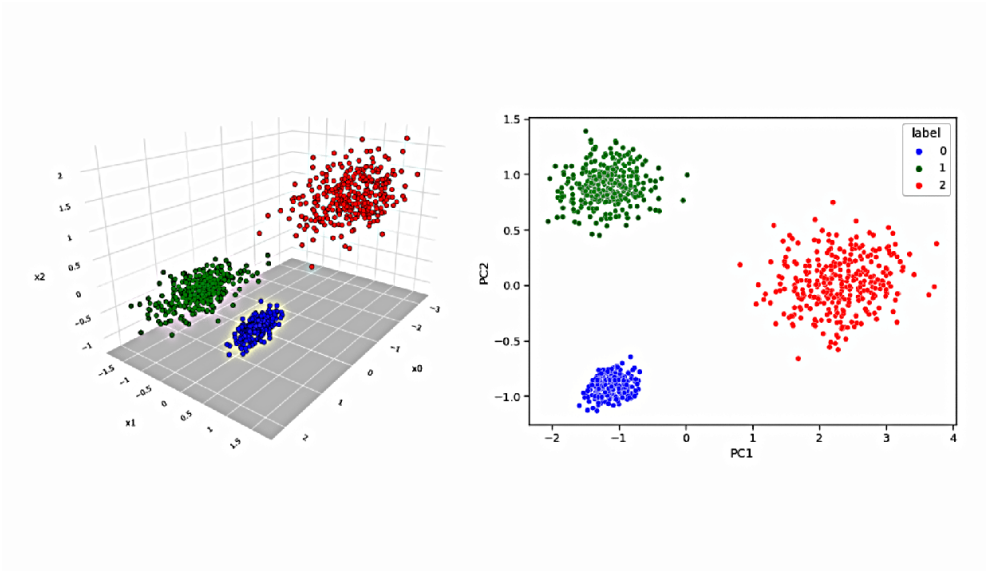


Fig. 7.1 Projection of NIFTY 500 constituents onto the first two Principal Components. Stocks (points) naturally form clusters corresponding to their industry sectors, verifying that the geometric distance in the reduced subspace preserves economic relationships.

7.4 Validation: The Residual Test

The last validation step is to check the residuals which is the information we remove during dimension reduction. Let A_k be the rank-k version of the market matrix A. The residual matrix $E = A - A_k$ stands for the noise.

If the model is working correctly, the rows of E should behave like a Gaussian Annulus (isotropic noise) that we talked about in Chapter 3. If we still see correlation or some pattern in E, then maybe the value of k was too small and we accidentally threw out some useful signal.

Chapter 8

Conclusion and Future Work

8.1 Conclusion of Phase 1

This report shows the end of the first phase of the project where we mainly tried to make a geometric way to deal with the Curse of Dimensionality in the Indian Stock Market. We have seen that the NIFTY 500 is too complex for normal statistical tools because the data is very sparse and noisy in high dimensions.

In Phase 1, we managed to do the following.

- **Framed the Problem Geometrically:** We shifted the analysis from a normal correlation based approach to a geometric one where market states are taken as vectors in Euclidean space.
- **Established Theoretical Bounds:** By using the Law of Large Numbers and the Gaussian Annulus Theorem, we showed that a market that is fully random would have strict distance concentration. This gives the mathematical baseline needed to catch non random market patterns.
- **Designed the Methodology:** We made a full framework using Singular Value Decomposition (SVD) and the Johnson Lindenstrauss Lemma to project the 500 dimensional market cloud into a smaller factor space without losing much information.

The theory we made here shows that the intrinsic dimension of the market is much smaller compared to the number of stocks. This supports the idea that dimensionality reduction is useful for proper financial modeling.

8.2 Plan for Phase 2

The second phase of the project will now move from theory to practical coding and testing. The main tasks planned for next semester are listed below.

1. **Implementation of the Data Pipeline:** We will build a Python system to collect, clean and normalize around 10 years of daily price data for the NIFTY 500. This also needs handling things like splits or dividends and making sure the log return vectors stay stationary.

2. **Subspace Extraction via SVD:** We will apply the factorization ideas from Phase 1 to the real data. This includes creating the eigenvalue spectrum and using Scree Plots to find the intrinsic dimension k.

3. **Factor Identification:** The remaining principal components will be compared with outside economic signals (like NIFTY Bank Index or USD INR rate) to understand the real world meaning of the geometric factors.

8.3 Future Research Directions

While Phase 2 is centred on dimension reduction and factor understanding, this project also opens some long term research ideas that go beyond the B.Tech scope.

- **Geometric Portfolio Construction:** Using the cleaned lower dimensional covariance matrix to build mean variance optimal portfolios. Removing the noise dimensions should make the portfolios more stable out of sample.

- **Crisis Prediction:** Studying how the intrinsic dimension k changes with time. A fast drop in dimensionality (where $k \to 1$) can be a sign of a market panic which maybe can work as an early warning for a financial crisis.

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